Set
Tue 8 Sep 2026

Homework 2 — Set identities

Session 2 · Sets & set operations

HOW TO ANSWER Every answer carries one sentence of justification. A value on its own is not a solution — say why it is right. You may discuss problems with anyone and may use AI tools, but you must be able to explain every line you submit.

TWO FORMATS, AND BOTH ARE REQUIRED EXACTLY **To prove an identity:** double inclusion. Take an **arbitrary** element of one side, show it lies in the other, then do the reverse. The word "arbitrary" must appear.
To disprove one: name your sets explicitly, compute **both** sides, and point at the element that is in one and not the other. A Venn diagram is not a proof and is not a disproof.

Part A · Notation and computation

1 NOTATION

Let $U = \{1, 2, \ldots, 12\}$, $A = \{\, x \in U : x \text{ is even} \,\}$ and $B = \{\, x \in U : x \text{ is a multiple of } 3 \,\}$.

a. Write A and B in roster notation.

b. Compute $A \cap B$, $A \cup B$, $A \setminus B$, $A \triangle B$ and B^c.

SOLUTION (a) $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{3, 6, 9, 12\}$.
(b) $A \cap B = \{6, 12\}$; $A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$; $A \setminus B = \{2, 4, 8, 10\}$; $A \triangle B = \{2, 3, 4, 8, 9, 10\}$; $B^c = \{1, 2, 4, 5, 7, 8, 10, 11\}$.

2 ∈ VS ⊆

Let $C = \{\, 1,\ \{1\},\ \{1, 2\},\ \varnothing \,\}$. Decide true or false, and in each case quote the definition that settles it.

a. $1 \in C$ (b) $\{1\} \in C$ (c) $\{1\} \subseteq C$ (d) $\{1, 2\} \subseteq C$ (e) $\varnothing \subseteq C$ (f) $|C| = 4$

SOLUTION (a) True — 1 is listed. (b) True — $\{1\}$ is listed as an element. (c) True — the only element of $\{1\}$ is 1, and $1 \in C$. (d) False — it would need $2 \in C$, and 2 is not an element of C. (e) True — vacuously, for every set. (f) True.

3 POWER SET

Let $A = \{a, b, c\}$.

a. List $\mathcal{P}(A)$ in full, organised by the size of the subset.

b. State $|\mathcal{P}(A)|$ and explain in one sentence why $|\mathcal{P}(A)| = 2^{|A|}$ for *any* finite A.

SOLUTION (a) $\varnothing$; $\{a\}, \{b\}, \{c\}$; $\{a, b\}, \{a, c\}, \{b, c\}$; $\{a, b, c\}$.
(b) $|\mathcal{P}(A)| = 8 = 2^3$. Building a subset means deciding, independently for each of the n elements, whether it is in or out — two choices n times, so 2^n subsets.

Part B · Six identities to prove

FOR EVERY PROBLEM IN THIS PART Prove by double inclusion — both directions are required. A proof that verifies the identity on one example is **not a proof**, however carefully the example is computed.

4 DISTRIBUTIVITY

Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ for all sets A, B, C.

SOLUTION ($\subseteq$) Let $x \in A \cup (B \cap C)$ be arbitrary. Then $x \in A$, or $x \in B \cap C$. If $x \in A$ then x is in both $A \cup B$ and $A \cup C$, so in their intersection. If $x \in B \cap C$ then $x \in B$ and $x \in C$, so again $x \in A \cup B$ and $x \in A \cup C$. Either way $x \in (A \cup B) \cap (A \cup C)$.
($\supseteq$) Let $x \in (A \cup B) \cap (A \cup C)$ be arbitrary, so $x \in A \cup B$ and $x \in A \cup C$. If $x \in A$ we are done. If not, then from the first $x \in B$ and from the second $x \in C$, so $x \in B \cap C$. Either way $x \in A \cup (B \cap C)$. ∎

5 DE MORGAN

Let $A, B \subseteq U$. Prove that $(A \cap B)^c = A^c \cup B^c$.

SOLUTION ($\subseteq$) Let $x \in (A \cap B)^c$ be arbitrary. Then $x \in U$ and $x \notin A \cap B$, i.e. it is not the case that both $x \in A$ and $x \in B$. So $x \notin A$ or $x \notin B$, giving $x \in A^c$ or $x \in B^c$, hence $x \in A^c \cup B^c$.
($\supseteq$) Let $x \in A^c \cup B^c$ be arbitrary. Then $x \notin A$ or $x \notin B$, so it is not the case that x lies in both, i.e. $x \notin A \cap B$. As $x \in U$, $x \in (A \cap B)^c$. ∎

6 DIFFERENCE

Prove that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ for all sets A, B, C.

SOLUTION ($\subseteq$) Let $x \in A \setminus (B \cup C)$ be arbitrary. Then $x \in A$ and $x \notin B \cup C$, so $x \notin B$ and $x \notin C$. Hence $x \in A \setminus B$ and $x \in A \setminus C$, so x is in the intersection.
($\supseteq$) Let $x \in (A \setminus B) \cap (A \setminus C)$ be arbitrary. Then $x \in A$, $x \notin B$ and $x \notin C$, so $x \notin B \cup C$, giving $x \in A \setminus (B \cup C)$. ∎

7 SYMMETRIC DIFFERENCE

Prove that $A \triangle B = (A \cup B) \setminus (A \cap B)$ for all sets A, B.

SOLUTION Recall $A \triangle B = (A \setminus B) \cup (B \setminus A)$.
($\subseteq$) Let $x \in A \triangle B$ be arbitrary, so $x \in A \setminus B$ or $x \in B \setminus A$. In the first case $x \in A \subseteq A \cup B$ and $x \notin B$, so $x \notin A \cap B$. The second case is symmetric. Either way $x \in (A \cup B) \setminus (A \cap B)$.
($\supseteq$) Let $x \in (A \cup B) \setminus (A \cap B)$ be arbitrary. Then x is in A or in B, but not in both. If $x \in A$ then $x \notin B$, so $x \in A \setminus B$; otherwise $x \in B$ and $x \notin A$, so $x \in B \setminus A$. Either way $x \in A \triangle B$. ■

8 SUBSETS

Prove that for all sets A, B: $A \subseteq B \iff A \cup B = B$.

SOLUTION ($\Rightarrow$) Suppose $A \subseteq B$. Certainly $B \subseteq A \cup B$. Conversely, let $x \in A \cup B$ be arbitrary: if $x \in B$ we are done, and if $x \in A$ then $x \in B$ by the hypothesis. So $A \cup B \subseteq B$, and the two sets are equal.
($\Leftarrow$) Suppose $A \cup B = B$. Let $x \in A$ be arbitrary. Then $x \in A \cup B$, which equals B, so $x \in B$. Hence $A \subseteq B$. ■

9 CARTESIAN PRODUCT

Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$ for all sets A, B, C.

SOLUTION Elements here are ordered pairs, so an arbitrary element is written (x, y).
($\subseteq$) Let $(x, y) \in A \times (B \cap C)$ be arbitrary. Then $x \in A$ and $y \in B \cap C$, so $y \in B$ and $y \in C$. Hence $(x, y) \in A \times B$ and $(x, y) \in A \times C$.
($\supseteq$) Let $(x, y) \in (A \times B) \cap (A \times C)$ be arbitrary. Then $x \in A$, $y \in B$ and $y \in C$, so $y \in B \cap C$ and $(x, y) \in A \times (B \cap C)$. ■

Part C · Three to disprove, and one to write

10 DISPROOF

Each statement below is **false**. For each, give explicit sets, compute both sides, and name an element that is in one side and not the other.

a. $(A \cup B) \setminus B = A$ for all sets A, B.

b. $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ for all sets A, B.

c. $A \times B = B \times A$ for all sets A, B.

SOLUTION (a) $A = B = \{1\}$: left is $\{1\} \setminus \{1\} = \varnothing$, right is $\{1\}$. The element 1 is in the right and not the left. (It holds precisely when $A \cap B = \varnothing$.)
(b) $A = \{1\}, B = \{2\}$: left is $\mathcal{P}(\{1, 2\})$, which contains $\{1, 2\}$; right is $\{\varnothing, \{1\}\} \cup \{\varnothing, \{2\}\}$, which does not. The element $\{1, 2\}$ separates them. (Only $\supseteq$ holds in general.)
(c) $A = \{1\}, B = \{2\}$: left is $\{(1, 2)\}$, right is $\{(2, 1)\}$, and $(1, 2) \neq (2, 1)$ since ordered pairs agree only when both coordinates do.

11 FIND THE ERROR

A student submits the following. Identify the first step that is wrong, say why it is wrong, and state whether the claimed identity is nevertheless true.

Claim. $A \setminus (B \setminus C) = (A \setminus B) \cup C$ for all sets A, B, C.
"Proof." Let $x \in A \setminus (B \setminus C)$. Then $x \in A$ and $x \notin B \setminus C$. Now $x \notin B \setminus C$ means $x \notin B$ or $x \in C$. So $x \in A \setminus B$ or $x \in C$, hence $x \in (A \setminus B) \cup C$. Therefore the two sets are equal. ■

SOLUTION The individual steps of the displayed argument are correct, and they establish $A \setminus (B \setminus C) \subseteq (A \setminus B) \cup C$. The error is the final sentence: only **one inclusion** was proved, and equality requires both. In fact the identity is **false** — take $A = \varnothing, C = \{1\}$: the left side is $\varnothing$ and the right side is $\{1\}$.

12 REFLECTION

In no more than five sentences: a Venn diagram convinced you that some identity was true, and then you were asked to prove it by double inclusion anyway. What does the proof give you that the picture does not? You are not being asked to prove anything here.

> **WHAT A GOOD ANSWER DOES** Look for the idea that the diagram shows one configuration while the claim quantifies over all sets, and/or that a diagram cannot be drawn faithfully for four or more sets. Credit any answer that distinguishes "I am convinced" from "it is established". There is no single correct answer.